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HAPPY IN THE SERVICE OF THE COMPANY: THE PURCHASING POWER OF VOC SALARIES AT THE CAPE IN THE 18TH CENTURY

Sophia Du Plessis and Stan Du Plessis¹

ABSTRACT

This paper contributes to the debate on the level and trajectory of welfare at the Cape of Good Hope during the 18th century. Recent scholarship (for example, Allen 2005) has calculated and compared the levels and evolution of real wages in various European and Asian economies since the early modern period. To this literature we add evidence for unskilled and skilled workers of the Dutch East India Company at the Cape of Good Hope during the 18th century, following De Zwart (2009; 2011), who recently presented evidence for unskilled workers in the Cape for the latter half of the 17th century and the 18th century. We calculate job-specific real wages in a three-step argument; from the narrowest international comparison of wage rates in terms of silver content to one based on a basket of widely consumed goods. This paper adds to this literature by adapting the consumption basket for local circumstances (due to both diet and relative prices) and the comparison for local demographics. We also provide a broader range of comparative statistics on real wages. Finally, we add the real wages of skilled workers to the comparison of unskilled workers offered in the literature to date. While the paper is based on real wages for VOC officials the mechanism we identify as the cause of this rising prosperity (sustained lower prices of consumption goods) would have raised the prosperity of all colonists at the Cape.

Keywords: real wages, Dutch East India Company (VOC), Cape Colony, comparisons of living standards, economic history of South Africa, economic history of the Cape Colony

JEL classification: N37, N97, E31

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1. INTRODUCTION

Despite the optimistic name, the Cape of Good Hope hardly suggested an attractive turn in the career of a Dutch East India Company official in the first decades since Van Riebeeck's founding of the Cape Colony. By around 1700, and after 48 years of the colony's existence, an official at the Cape lived worse materially, not to mention culturally, than his counterparts back home in the Netherlands or in the Eastern hub of Batavia (Giliomee 2004). But as the fortunes of the VOC declined internationally during the course of the 18th century, its officials at the Cape enjoyed a happy rise in their living standards relative to their peers in Holland. We trace that rise in real wages and show that the underlying cause was not a change in the remuneration of VOC officials, but a dramatic change in what their salaries could buy. We observe a sharp rise in real wages as prices declined for essential consumption goods along with the slowly expanding market economy at the Cape.

The Cape Colony was founded in 1652 by the VOC to serve as a refreshment station for the trade route between Europe and the Indies. As the native Khoi could at best supply some meat for the passing ships, Governor Jan van Riebeeck allowed a number of VOC employees to grow wheat and (later) wine on the slopes of Table Mountain, with agriculture later to spread towards the fertile agricultural land west of the Drakenstein and Hottentots Holland mountains. Despite the geographical expansion, agricultural production increased only slowly, with several decades passing before the colony could feed itself (Ross 1986). Further expansion was initially checked by the prohibitive transport costs across these mountain ranges.

The climate, which favours European crops such as wheat and wine, and the distance from European markets, prevented the Cape of Good Hope from becoming a booming export colony in the 17th century. This is not to claim that exports were non-existent; there were notable opportunities to export European products to European communities in the tropics (Ross 1986), and wine elsewhere in the Southern hemisphere (Van Zyl 1975) where geography gave the cost advantage to the Cape, but these trade routes did not allow for an export-led boom (Giliomee 2004). This left the local market, comprising the urban centre at Cape Town, where most of the proportionally large group of VOC employees lived and worked, where non-agrarian free burgers plied various trades, and where a large number of urban slaves laboured.

A service economy emerged in Cape Town that not only catered for the circulating population of sailors, but also provided maintenance and supplies to the ships for the next leg of their journeys. This economy included, *inter alia*, taverns, lodging, shops, bakers, brewers, builders and carpenters (for residential and ship maintenance), smiths, coopers and potters (Ross 1989). The extent of the service economy is easily seen in the 1732 survey by Governor Jan de la Fontaine

which found no less than 60% of the population in Cape Town active in the service industry (Boshoff and Fourie 2010).

A number of economic historians, for example De Kiewiet (1941) and more recently Feinstein (2005), treated the early colonial history of the South African economy in a somewhat cursory manner, and focused their attention on the dramatic transformation of the economy following the discovery of vast diamond and gold reserves in the second half of the 19th century. While we choose to dispute their account the difference is one of emphasis; the frontiersman of the Colony were poor and their local economy stagnant (De Kock 1924) and this is where much of the historiographical attention has fallen. But there is also evidence of wealth, especially in and around Cape Town (Ross 1999). And Van Duin and Ross (1987) provided the statistical foundations for a description of an expanding and ever more productive economy at the Cape during the 18th century. However, their account is based on agricultural output alone, to the exclusion of the large services sector at the Cape.

It is relatively harder to reconstruct the productivity and total value added of the services sector at the Cape compared with the task of recording the physical output of the agricultural sector. But the impact of higher productivity in services can be inferred from existing data along two lines: firstly, by examining the stock of wealth accumulated by local colonists and secondly by comparing the flow of real income they received annually. The first approach was taken by Fourie (2012) who used probate inventories to highlight both the remarkable wealth of the elite colonists at the Cape as well as the emergence of significant wealth inequality. The second approach was taken by De Zwart (2009, 2011) who offered internationally comparable real wage calculations for VOC officials at the Cape during the 17th and 18th centuries. Finally Fourie and Van Zanden (2012) provided a flow calculation, based not on real wages, but on real GDP for the Cape colony.

This paper adds to this literature by adapting the consumption basket for local circumstances (due to both diet and relative prices) and local demographics. We also provide a broader range of comparative statistics on real wages. Finally, we add the real wages of skilled workers to the comparison of unskilled workers offered in the literature to date. While the paper is based on real wages for VOC officials, the mechanism we identify as the cause of this rising prosperity (sustained lower prices of consumption goods) would have raised the prosperity of all colonists at the Cape.

2. DATA SOURCES

Three sources supplied the bulk of the data used in this paper. Firstly, price data for meat, wine and grain were calculated from the appendices in Van Duin and Ross (1987, Appendix 10, Table 2), while prices for beans, butter, meat, bread, soap, candles, lamp oil, cotton and fuel were taken from the *Scheepssoldijboeken*

(Nationaal Archief 2008). The ‘Scheepssoldijboeken’ are available in the VOC archive (at the National Archive) in The Hague.² This resource documents the administration of ships passing Cape Town during the 18th century, but also the goods that they purchased in the Cape.

Thirdly, the TEPC (Transcription of Estate Papers of the Cape of Good Hope) (TEPC 2008) made documents on auctions, wages and inventories available in pdf format, from which we gathered both wage and price data. The originals are available in the Cape Town Archives Repository and were digitised in a joint project of the Universities of the Western Cape and Cape Town in partnership with the Cape Town Archives Repository and the National Archives at The Hague.

Salaries for VOC officials at the Cape were recorded in muster rolls, now held at the Cape Archives Repository, and the Nationaal Archief in The Hague. For this project, the wage data on the muster rolls were central, which left the large number of muster rolls from the Cape Archives Repository, where wage data is not available, out of consideration. However, the muster rolls from the Nationaal Archief do record wage data, and the records we used are listed in Appendix 1.

From the muster rolls, we recovered 4518 useable entries, comprising data from 1699 (Muster roll NA12653) to 1790 (Muster roll NA 12602). These muster rolls contain information on the specific occupation for which the VOC paid a monthly wage (‘gagie’ in guilders per month), which allowed us to arrange the data in eight sectors: Administration, Agriculture, Construction, Education, Church, Merchants, the Legal profession, Medicine, the Military, Shipping, Trades, and Transport. A brief discussion of the wage data (available electronically from the authors) is given in Appendix 1.

Of course, wages paid by the VOC might not reflect the cost of comparable labour on the open market, and these calculations, accordingly, do not present a reliable approximation of living standards for the colony as a whole. De Zwart (2009) identifies two reasons that suggest the possible underestimation of real wages when using muster roll data: first the company paid its lowest officials (soldiers, who were also employed in some low-skilled tasks) less than the going wage rate for low-skilled workers amongst free burgers.³ Second, company officials received a part of their remuneration in goods and services,⁴ such as company housing, though the value of these additional resources remains uncertain. De Zwart (2009) concludes reasonably that real wage calculations based on the muster rolls offer a lower bound to the actual real wage at the Cape for this period.

2 The authors are grateful to Pim de Zwart who used the data from the Scheepssoldijboeken in De Zwart (2011) and generously made it available for this study.

3 De Zwart (2009) compared the salaries of soldiers on the muster rolls with the wage a free burger could earn in construction given by Biewenga 1999.

4 See especially Schutte 1989 for a discussion of these additional benefits.

We compiled price series for grain and wine from the inventories, and prices for auctions held on the estates of the deceased. The series of auction lists used here (MOOC10) is available in pdf format in the TEPC database (TEPC 2008).

While nominal wages reveal the primary distribution between employees in the different occupations, they do not indicate what bundle of goods and services the salaried worker could buy; in other words, they do not identify standards of living. To calculate changes in living standards, we needed to calculate real wages, to which end we constructed a total price index from the prices of goods that would have accounted for substantial proportions of the regular expenditure of a household in the 18th century. Table 2.1 in Appendix 2 lists the sources, original units of measurement and conversion formulae to guilders, kilograms and litres, while the time series data is available electronically from the authors. A discussion of each price series is also provided in Appendix 2.

3. LIVING STANDARDS IN AN INTERNATIONAL COMPARISON

By combining the nominal wage series compiled above with the prices of goods consumed by typical households, it is possible to compare the living standards of families with breadwinners from particular occupations with families in different circumstances, both at the Cape and internationally. Such comparisons can be approached in a number of ways, three of which have been implemented here, starting with two simple comparisons and ending with a more elaborate comparison based on a basket of consumer goods. The calculations offered in this section extend De Zwart's (2009) comparison of standards of living at the Cape Colony from the mid-17th century to the early 18th century and his more recent calculations for the 18th century (De Zwart 2011).

3.1 A first comparison: silver wages

Expressing the local wage in terms of an internationally comparable standard of value is an obvious way to proceed with the international comparison of standards of living. For the period under consideration, this means expressing the local wage in terms of silver so that it can be compared with silver wages elsewhere in the world at that time. This calculation presents no difficulties for the Cape Colony during this period as guilder wages have already been established, and the silver content of guilders remained constant (at 9.61 grams of silver per guilder) from 1681 onwards for the remainder of the period under consideration.⁵

5 The silver value of guilders are recorded as a part of the extensive "Historical Wages and Prices" database constructed by Van Zanden at the Institute of Social History in Utrecht and available online at: <http://www.iisg.nl/hpw/brenv.xls>.

Table 1: Comparison of unskilled wages using silver content

Period	Source	Location	Daily wage in grams of silver
1730	Allen et al. (2005)	London	10
1790	Allen et al. (2005)	London	11
1650–1699	Broadberry and Gupta (2006)	Southern English towns	5.6
1730	Allen et al. (2005)	Amsterdam	8.5
1790	Allen et al. (2005)	Amsterdam	9
1874	Broadberry and Gupta (2006)	Northern and Western India	1.79
1790	Broadberry and Gupta (2006)	Southern India	1.44
1730	This paper	Cape Town	3.5
1790	This paper	Cape Town	5

The unskilled worker in this calculation is a soldier at the Cape, an assumption also made by De Zwart (2009).

To complete this calculation, an assumption about the number of workdays per month is required to turn monthly guilders into an internationally comparable silver wage per day. Following De Zwart (2009), we assumed a total of 300 workdays per year, or 25 workdays per month.⁶ The results of this comparison for an unskilled worker are shown in Table 1, where the daily silver wage for a soldier (here assumed to be an unskilled worker) at the Cape is compared with those of unskilled workers in leading economic centres of the day (London and Amsterdam) as well as rural England and elsewhere in the greater Indian Ocean area (Northern and Western India and Southern India).

The comparison of silver wages suggests that unskilled labourers lagged significantly behind their peers in the leading economic centres of North-Western Europe at the start of the 18th century, a result consistent with De Zwart's (2009) calculations for the latter half of the 17th century. The comparison is more favourable with other cities in North-Western Europe and with towns in Southern England outside of London and comfortably ahead of unskilled wages in India during the 18th century and even during the late 19th century. More importantly, while wages expressed in silver remained largely constant in North-Western Europe⁷ over this period, they increased at the Cape, suggesting that workers at

6 This assumption is based on Allen's (2001) calculation of 300 workdays per year for a Protestant society in Northwest Europe at this time.

7 London is an exception, with silver wages rising fast after 1790 (Allen 2005).

the Cape were catching up with their more affluent peers in Europe during the course of the 18th century.

Silver wages are not, however, an adequate basis for comparing the purchasing power of wages internationally. The *law of one price*, which (after adjusting for transport costs) equalises the cost of goods and services internationally, does its work over a horizon of decades, and at any point in time, local prices can and do diverge greatly from prices elsewhere, even with extensive international trade. Because the same gram of silver bought a very different basket of goods in Cape Town and in Amsterdam, a more accurate estimate of the relative real wage in these cities can only be achieved by scaling the nominal wage by goods and services actually bought and consumed.

3.2 A second comparison: the price of grain

Parthasarathi (1998) took a methodologically simple approach to international comparisons of purchasing power by comparing the amount of grain a salary could purchase in local markets. Using this method, the nominal wage is expressed as equivalent to a certain number of pounds of grain per week; a ratio can then be compared across societies. Parthasarathi (1998) reported these calculations for Britain and India in the 18th century. We use it here to compare the earnings of VOC employees at the Cape and similarly employed workers in Northern Europe. Table 2 shows the relevant results for Britain as calculated by Parthasarathi (1998) for the middle of the 18th century (for lower and higher skilled weavers) and calculations for (lower skilled) soldiers and (higher skilled) carpenters at the Cape.

From Table 2, it appears that a low-skilled VOC employee had more than caught up with a low-skilled English weaver by the middle of the 18th century, and more prosperity was to follow. Skilled employees were further behind their English peers in this comparison, though they too were catching up notably by mid-century.

Table 2: Purchasing power of real wages in grain

Period	Source	Occupation	Grain per week (pounds)
Mid 18th century	Parthasarathi (1998)	Rural English weaver	40–50 pounds
Mid 18th century	Parthasarathi (1998)	Spitalfields	120–140 pounds
		Silkweavers	
Early 18th century	This paper	Soldier	42 pounds
Mid 18th century	This paper	Soldier	55 pounds
Late 18th century	This paper	Soldier	70 pounds
Early 18th century	This paper	Carpenter	74 pounds
Mid 18th century	This paper	Carpenter	86 pounds

But this calculation is also inadequate; the method exaggerates the importance of one price – that of the chosen staple – to the exclusion of all other information about the economy. For example, inventories that survived show a rich diet even in the household of a person like Joachim von Dessin, who arrived in the Cape as a VOC soldier in 1727. The thorough record of his household includes an inventory of his scullery, which included, *inter alia*, the following: wheat, flour, rice, sago; a variety of beans (yellow, white, black, brown); spices, such as mustard, cumin, pepper and tamarind; sugar and coffee beans; oil and lard; meat (beef and pork); Dutch wine and wine from the Cape, as well as soap, wood and candles (Claassens 2006, 118). Since the cost of living is inherently multi-dimensional, reducing the comparison of real wages to a comparison of grain purchasing power risks distorting the international comparison, as Allen has argued elsewhere (Allen 2001).

3.3 A third comparison: the cost of living

De Zwart recently found that living standards at the Cape during the latter half of the 17th century were lower than in the Netherlands, but not out of line with the experience of the rest of North-Western Europe, and higher than elsewhere in Europe or in Asia (De Zwart 2009). Using very detailed price data from VOC correspondence in the National Archive, he was able to calculate the silver price of three consumption baskets at the beginning of the 17th century. This is part of a larger body of international literature that compares standards of living across both time and societies by expressing nominal income relative to the cost of consumption baskets that attain certain levels of nutrition expressed in calories and protein. For example, Allen et al. (2011) propose two different baskets for the sake of such comparisons: a “barebones” basket which gives 1946 calories and 82 grams of protein per day, together with minimum requirements in terms of fuel, textiles and so on, while the calorie content of their “respectable” basket is higher at 2140 per day, and the other components of the basket are also adjusted upwards.

De Zwart (2011) based his welfare comparisons on baskets as close as possible to that of Allen et al. (2005). In this paper, we replicate his results and suggest some variations on the baskets, adjusted for local circumstances. Table 3 shows the “bare bones” baskets of Allen et al. (2011) compared with De Zwart’s (2011) “bare bones” basket and two baskets proposed in this paper. The differences between the baskets of De Zwart and Allen et al. lie in De Zwart’s use of grain (instead of bread) as the staple and the higher consumption of butter in the De Zwart basket. There is little to choose between the nutritional value of either basket, with De Zwart’s giving 1945 calories and 89 grams of protein per day in contrast to the 1946 calories and 82 grams per day of Allen et al.’s basket.

The decision by De Zwart to raise the consumption of butter relative the Northern-European basket of Allen et al. is justified by the known higher dairy

Table 3: Annual composition of “bare bones” baskets

Item	Unit	Allen et. al	De Zwart	Basket 1	Basket 2
Grain	Kg		173		
Bread	Kg	247*		223	variable
Beans	Kg	20	20	20	20
Meat	Kg	5	5	26	variable
Butter	Kg	3	5	4	3
Soap	Kg	1.3	1.3	1.3	1.3
Candles	Kg	1.3	1.3	1.3	1.3
Lamp oil	Litre	1.3	1.3	1.3	1.3
Cotton	m ²	3	3	3	3
Fuel	MBTU	3	3	3	3
Calories per day		1936	1945	1939	
Grams of protein per day		82	89	78	

Sources: Allen et al. 2011; De Zwart 2011; and own calculations.

*Allen et al. (2011) proposed a “bare bones” basket with 155kg of oats per year that would yield 1657 calories per day. We used the calorie content per kilogram of bread listed in Allen et al. (2011, Table A2), i.e. 2450 calories per kg to derive the amount of bread, 247kg per year, that would also yield 1657 calories per day.

content of diets at the Cape. Following the same logic, the proposed baskets in this paper adjust the meat content of the local diet upwards at the cost of the staple consumption. We followed De Zwart in raising butter consumption (though not by as much) and Allen et al. in preferring bread to grain as the appropriate staple. “Basket 1” proposed here gives a comparable 1939 calories and 78 grams of protein per day.

There are good reasons for raising the meat content of the local baskets. First, as with dairy, commentators on the colonial diet at the Cape emphasised the importance of meat (for example, Botha 1962). More persuasively from an economic perspective, the relative price of meat at the Cape encouraged meat consumption relative to bread. At the start of the 18th century, for example, 1 kg of meat cost 2.27 times as much as 1 kg of bread in Amsterdam,⁸ a ratio that was 2.75 at the Cape. As the 18th century unfolded though, the price of meat rose dramatically in Holland, while it declined sharply at the Cape. By 1793, the price ratio between meat and bread was 3.65 in Amsterdam and only 0.82 at the Cape.

The second basket proposed in this paper is based more explicitly on the time variation in the relative price of meat and bread over the course of the 18th century. This “economic” basket is the same as the Allen et al. “bare bones”

8 The price of meat and bread in Amsterdam is from Van Zanden 2008.

basket for Northern Europe, but allows for time-dependent combinations of bread and meat. In each year, “basket 2” is the cheapest combination of bread and meat that would satisfy minimum calorie requirements (1691 per day) as well as give no less than 70 grams of protein from bread and meat combined, subject to a minimum bread consumption of 150 kg per year.⁹

The evolution of these “barebones” baskets over the 18th century given the price series discussed above is shown in Figure 1, after scaling each to an index value of 100 for the year 1700.

In all four cases, a substantial decline – in the order of 15% – in the nominal cost of a “bare bones” basket is observed over the period under consideration. Basket 1 showed the largest decline (29%) due to the larger meat consumption, the price of which declined faster than the prices of most other large components of these baskets.

While the “bare bones” basket provides for a minimum level of nutrition, a higher income would allow a person substantially greater nutritional value. Allen et al. (2011) proposed a “respectable” consumption basket to capture this higher potential standard of living. Their “respectable” basket, as well as a variant responsive to the different relative prices of meat and bread (basket 3), was used as benchmarks in this paper. Table 4 shows the annual composition across products for the two “respectable” baskets used in this paper.

The evolution of the two “respectable” baskets given the price series is shown in Figure 2, scaled to 100 in 1700. While the cumulative declines in the nominal value of these baskets are slightly less than for the “barebones” baskets, the declines are substantial: 15% for the Allen et al. basket and 20% for basket 3.

The ratio of a nominal wage series to the nominal cost of a consumption basket yields what, in this literature, has been called a welfare ratio. When it exceeds 1, the ratio indicates that the wage is sufficient to cover the expenses associated with the consumption basket. Our interest here lies with the relative prosperity of a low-skilled worker and a higher-skilled worker at the Cape during the 18th century in an international comparison. To that end, we chose a VOC soldier as being representative of a low-skilled worker, for whom we had extensive wage data, and a VOC clerk (or assistant) as an example of a higher-skilled worker.¹⁰

9 As stated, the rules of the second basket exhibit a linear programming problem that yields one of two baskets in any particular year. When meat is relatively expensive, the basket comprises 247 kg of bread and 5 kg of meat per year and switches to a basket with 150 kg of bread and 100 kg of meat when meat is relatively cheap.

10 The muster rolls contained nominal wage data for assistants until 1756, but with very little variation. A wage of 20 guilders per month seems to have been the going wage at this level during the 18th century, and for the purposes of this paper, was assumed to hold until 1793. De Zwart (2011) also observed the small variation in nominal wages, and likewise, assumed that nominal wages remained constant for much of the century.



Figure 1: “Barebones” consumption indices for the Cape during the 18th century

However, a number of adjustments are required before the nominal wage series from the muster rolls can be compared with the consumption baskets to calculate welfare ratios. First, the consumption basket has to be adjusted upwards

Table 4: Annual composition of “respectable” baskets

Item	Unit	Allen et. al.	Basket 3
Grain	Kg		
Bread	Kg	247	168
Beans	Kg	20	20
Meat	Kg	5	45
Butter	Kg	3	3
Soap	Kg	1.3	1.3
Candles	Kg	1.3	1.3
Lamp oil	Litre	1.3	1.3
Cotton	m ²	3	3
Fuel	MBTU	3	3
Wine	Litres	70	70
Calories per day		2140	2140
Grams of protein per day		78	78

Sources: Allen et al. (2005) and own calculations.

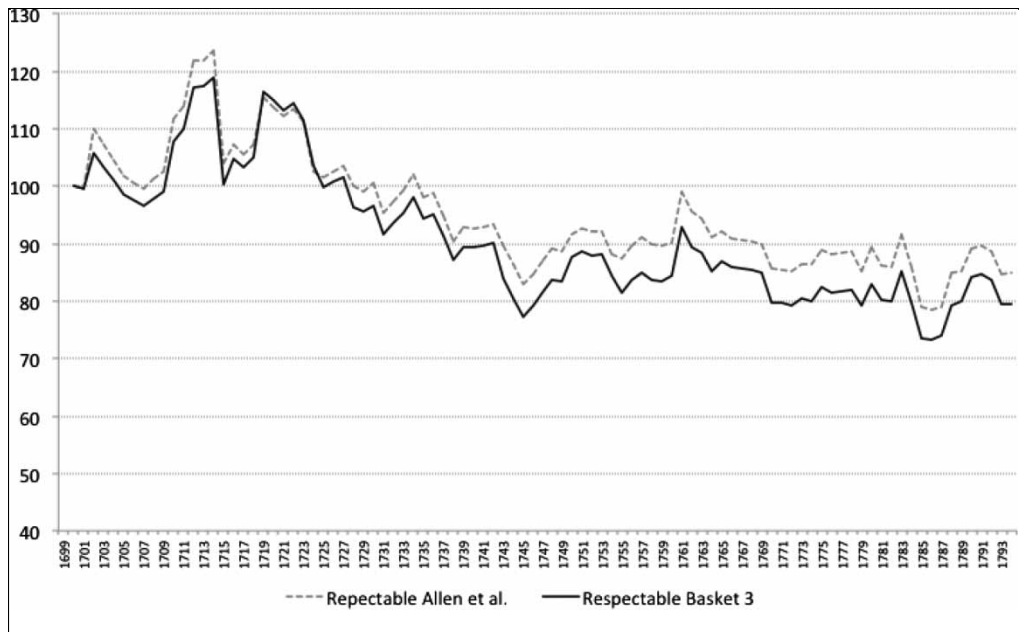


Figure 2: "Respectable" consumption indices for the Cape during the 18th century

to include housing costs. Following Allen et al. (2011) as well as De Zwart (2011), a rental cost of 5% was added to the consumption basket. Second, a married worker's wage might support a family, and to allow for the family De Zwart (2011) followed Allen et al. (2011) to assume that three times the consumption basket would yield the nutritional content required for a family of two adults and two small children. While this adjustment is conventional it might not be appropriate for the Cape during the period under consideration, at least during the earlier decades of the 18th century, on account of the relatively small female population of the colony at that time. Opportunities to marry and start a family would have been limited by this demographic imbalance. Fourie and Van Zanden (2012), for example, record only 168 women and 238 children compared with 394 men in the settler population of 1692, numbers which rise to 830 women, 2384 children and 1244 men in 1748 and further to 2870 women, 7800 children and 4259 men in 1795 (Ross 1975). Migration remained an important source of population growth throughout the century and immigration was largely male. The romantic prospects for company officials would, accordingly, have been limited, at least for conventional relationships in what was considered a "Christian society" (Ross 1999). On the other hand, the data confirms the impression of large families for those who did get married, a result of "early marriage, frequent childbearing and a relatively long period of fertility" (Ross 1975, 228).

Given these circumstances we assumed a smaller family size for the average VOC official. The median ratio of women to men in Ross (1975) is 67.3 with a standard deviation of only 4.7. Against this the relatively large number of children would raise the nutritional requirements of families. To balance the impact of

diminished opportunities for marriage with the impact of larger families for those who did get married we assumed that the average family for a VOC official at the Cape was 70% of the size assumed by Allen et al. (2011), i.e. a family that required 2.1 times the nutrition of a mature man.

Third, De Zwart (2009) correctly observes that the VOC paid at least some part of a worker's remuneration in kind, though such payments were not recorded in the muster rolls, and there are no reliable records of their size. A typical example of these benefits include the (comfortable) house and kitchen garden which was included in the benefits received by R.S. Alleman when he became the master of the Company's stable in the 1720s. Mentzel (1784), Allemans' biographer, also mentions the additional sources of revenue open to an official at the stables, including permission to trade in tobacco and pipes and even in wine (Mentzel 1784, 45–6). While Allemans had permission to trade for his own account in these cases, there was considerable scope for Company officials to do so without permission. These illegal activities by company official included agriculture and trading in goods and services (Fourie 2012). Both legal and illegal sources of income would have raised the welfare of Company officials above the levels calculated here.

A welfare ratio for soldiers is shown in Figure 3, where the nominal wage series was compared with the four “barebones” baskets. Figure 4 shows a welfare ratio for clerks, where the nominal wage series was compared with the two “respectable” baskets.

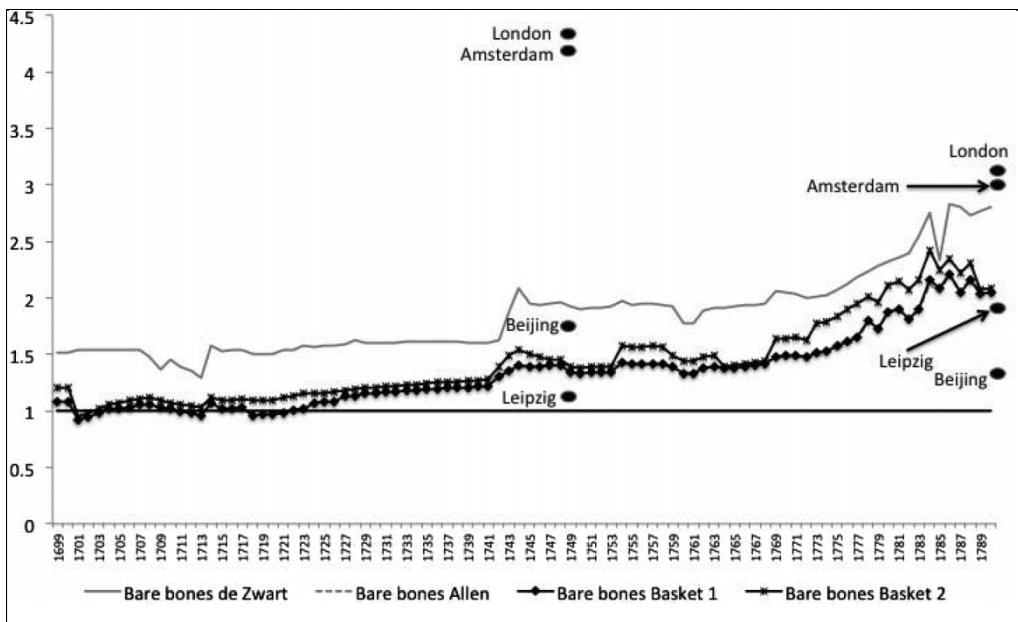


Figure 3: Welfare ratios for soldiers at the Cape during the 18th century
Source: own calculations and data from Allen et al. 2011.

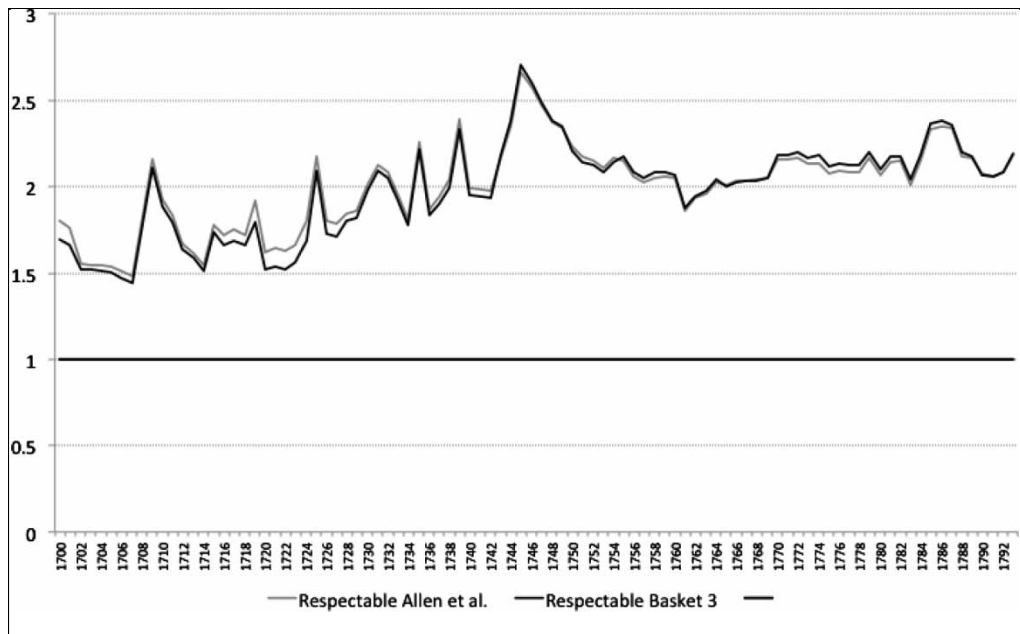


Figure 4: Welfare ratios for clerks at the Cape during the 18th century

The welfare ratios in Figure 3 confirm De Zwart's (2011) results. At the start of the 18th century soldiers had a welfare ratio in the order of 1.5, which rose steadily to 2.8 using De Zwart's "bare bones" basket. These results are closely comparable to his results both in magnitude and trend. The welfare ratios based on bread (as opposed to grain) start at a lower level, with a value of 1 for the newly constructed basket 1 and a value of 1.2 for both the Allen et al. basket and the newly constructed basket 2. These baskets based on bread rise strongly over the course of the century to reach a value of 2.1 at the end of the period.

At the beginning of the 18th century, a VOC soldier's wage would have supported a family at a fairly low level of welfare by comparison with the cost of various "bare bones" baskets and also in an international comparison. Workers were better off on this measure in cities such as Amsterdam and London and even Beijing, though not in Leipzig. Despite stagnant nominal wages, the decline in the nominal cost of living at the Cape during the 18th century raised welfare substantially towards the end of the century. Welfare for VOC workers at the Cape rose to a degree comparable to that of emergent centres of affluence in Europe such as Leipzig over this period, overtaking the welfare of workers in Beijing and substantially closing the gap on workers in Amsterdam and London where real wages had been declining (Allen et al. 2011; De Zwart 2011).

Figure 4 shows the welfare ratio for a more skilled VOC employee at the Cape. Such an employee started the century in a much better position, even when

compared with the cost of the “respectable” basket. Again, the decline in the cost of living raised the welfare ratio of assistants by 20–30%.

The welfare ratios, silver content and grain purchasing power comparisons reported here confirm the work of De Zwart (2011) on the increased welfare of unskilled labour at the Cape in the 18th century. Skilled workers started the century at a higher level of welfare (even compared with workers linked to a more expensive consumption basket), and they also experienced a steady gain in welfare as the century unfolded. The few highest-paid company officials, such as the governor and the ministers of religion in Cape Town and Stellenbosch were already affluent by international standards as the century opened and saw their welfare rise steadily with the passing years. The vicar in Stellenbosch, for example, earned a salary more than nine times the cost of Allen et al.’s (2005) respectable basket in 1700. By the end of the century, the incumbent in that post, Reverend Borchers, could buy the respectable baskets almost 18 times over every year (his salary was rerecorded by Borchers 1861).

The mechanism which caused these welfare increases, a large and sustained decline in the price of important consumption goods, is consistent with Van Duin and Ross’s (1987) earlier argument that the colonial economy showed steady production growth over the course of that century. By this mechanism, VOC officials enjoyed increasing prosperity, and the same trend would have been true of free burgers not on the Company’s payroll.

This interpretation informs the related question as to why the VOC allowed the real wages of its officials to rise so significantly in this colony, while the commercial fortunes of the Company suffered a notable decline. An adequate answer to this question requires an investigation into the recruitment policies and methods of the company as well as a comparative study of the prosperity of officials in the Company’s various colonies. Though such an investigation falls outside the scope of this paper it is possible to point to the local origins of prosperity at the Cape, i.e. the favourable climate which facilitated productive agriculture, the providential arrival of immigrants with productive skills¹¹ and, at least occasionally, good governance.¹² These local factors imply that rising prosperity at the Cape was not the Company’s explicit policy, but rather the unintended consequences of corporate decisions as well as the good fortune of geography.

11 See for example Fourie and Von Fintel’s (2011) demonstration of the impact by Huguenots from wine growing regions in France on the productivity of wine production at the Cape in this period.

12 The 18th century biographer Mentzel, described the Cape as “... indisputably the best place in the whole East Indies for military service” (Mentzel 1784, 12). The reason Mentzel gave for this favourable assessment had much to do with governance at the Cape, i.e. the formal conditions of employment for soldiers and the treatment they received as employees.

4. CONCLUSION

This paper is the outcome of a study of the comparative welfare of VOC officials at the Cape during the 18th century. It involves combining data on nominal wages and the prices of consumption goods. At the beginning of this period, a worker at the Cape would have experienced a comparatively low standard of living in an international comparison, a position that would have improved considerably over the course of that century. While nominal wages hardly moved, real wages rose sharply as prices declined. The position of a skilled worker was more favourable at the start of the period under consideration and also improved steadily over the course of the century.

These results have two implications: first, falling prices in the context of steadily expanding production suggests an expanding and more productive market economy, not over-supply, which would have led to contracting production over time.¹³ To put it differently, we have provided further evidence of a slowly but steadily expanding market economy at the Cape during the 18th century. Second, to the extent that rising real wages for VOC officials derived from falling prices (and not rising nominal wages), the opportunity for a higher standard of living would have been spread to the entire population, those who were happily employed by the company and the (perhaps even happier) free burgers.

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13 In a competitive market the rational response to market signals leads from oversupply to a subsequent contraction in production and increased production in response to excess demand. This adjustment mechanism is undermined when competition is compromised and the VOC's monopsonistic pricing policy would likely have affected market adjustments. But the VOC's ability to fix the market was limited and the ability of producers to trade illegally and of officials to trade for their own account would have improved the competitiveness and the feedback mechanism of the market for goods and service in the Cape.

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APPENDIX 1

A brief discussion of the wage data, grouped by sector of economic activity is given below.

Muster rolls from the Nationaal Archief, The Hague, used for wage data

Contents	Archival Code	Period
Copia monsterrolle vant generaale guarnisoen danno 1699–1700: voor Zeeland N:o 43	12653	1699–1700
Generaale Monsterrolle des Casteels de goede Hoop onder ult:o Junij 1724 voor Zeeland	12537	1724
Generaale Monster-rolle des Casteels de Goede Hoop onder ultimo Junij 1726 voor de Caamer Zeeland	12539	1726
Vrij boek der Europiaanse Ingeseetenen des Casteels de goede Hoop d' a:o 17 4/45 van d' Camer Zeeland voor Zeeland per Spanderwou	12630	1745
Per't schip Middelburg Generaale Monsterrolle des Casteels de Goede Hoop onder ult:mo Junij 1756 voor Zeelandt	12561	1756
Monsterrol van alle de Militairen het Guarnisoen houdende Battallion van Cabo de Goede Hoop uijtmaakende onder Ult:o Junij 1772 voor Zeeland	12601	1772
Monster Roll van alle de Militairen het Guarnisoen houdende Bataillon onder ult: Juny 1790 uitmakende voor de Camer Zeeland	12602	1790

Administration

We classified 352 entries in the database as being data applicable to employees in the VOC administration, with the first entries relating to 1699 and the last relating to 1756. Included in this group are administrators overseeing the following: 158 assistants (adsistent), 15 masters (baas), 65 book keepers (boekhouder), 3

governors of the Cape of Good Hope, 5 classified as help (for instance, in the kitchen, in the weapons room) (handlanger), 3 stewards, 7 mandoores (these are slaves who acted as foremen to other slaves), 2 caretakers (oppasser), 1 wage master (pookagiemeester), 73 company representatives in the countryside (posthouders), 4 writers (schrijvers), and 11 signallers (vlaggeluif).

Agriculture

The primary sector consists of 158 entries in our database, with the first entries relating to 1699 and the last relating to 1756. The tasks of these workers included the following: 6 were cattle-herders (beestewagters), 4 were grass cutters (grasmaijers), 2 were dairy farmers (melkboer), 5 were reed-cutters (riet kappers), 15 were rangers (veldwagters), 9 were hunters (wildschutters), 25 were gardeners (thuijniers) (including 12 head gardeners), and 87 were lumberjacks (houtkappers) (including 7 head woodcutters, and 3 assistant woodcutters).

Construction

The construction sector consists of 214 entries in our database. The first entries are for 1699 and the last for 1756, with the following distribution across occupations: 4 head carpenters (timmerlieden), 50 house carpenters, 9 ship carpenters, 4 assistant carpenters, 6 turners (draaijer), 1 sculptor (beeldhouwer), 25 workers fortifying walls (fortificatiewerkers), 1 head stonebreaker and 10 stonebreakers (klippebrekers), 43 bricklayers (including 2 master messelaars and a messelaars assistant).

Education

There were only three teachers in the database for the first half of the 18th century, Adolf Hofman (earning a wage of 10 guilders per month in 1724 and 1726), Johannes Soigne (who was paid 9 guilders per month in 1724 and 1726) and Gotlieb Ramst, with a much higher 20 guilders per month in 1756. These wages are strikingly low in absolute and relative terms. By comparison, lumberjacks and gardeners regularly received 14 guilders per month.

The church

Sixty-seven VOC employees in our database served the Church in some capacity during the sample period, including 3 sextants, 1 organist, 1 independent minister (independend predicant), 30 ministers (predicant), 9 readers (voorlezer), and 22 sick comforters (krankbesoekers) serving Cape Town, Stellenbosch, the Swartland and Drakenstein.

Merchants

By the nature of the VOC's business, merchants were important company officials (distinguished as 'qualified persons' on the muster rolls), and our records include 75 merchants (koopman), of which 53 were assistant merchants and 9 were head merchants.¹⁴ There is no evidence of a trend in the nominal wage of any of these three categories over time though.

The legal profession

There are 217 entries for the legal profession in our database, including 117 assistants (in the police, the secretary of justice, the orphanage and the warehouse), 13 messengers (bodes), 44 book keepers (boekhouder), 3 jailers (cipier), 19 constables, 1 corporal, 1 fiscal (fiscaal), 4 military police (geweldigere), 1 steward of the court (hofmeester), 8 writers (schrijver), and 5 substitute magistrates (subsistut landdrost).

Medicine

For the medical sector, we classified 50 entries in our database, including 5 chemists (apotheker), 9 surgeons (chirurgijn, of which one was a master and 3 assistants), 25 masters of medicine in various positions of the hierarchy (oppermeester, tweede meester, derde meester, etc.), 7 persons who served as doctors (3 ziektroosters and 4 ziekvaders, of which 2 were provisional).

The military

The military sector is the largest sector in the database, consisting of 2538 data points, covering the longest period (up to 1790). Within this sector, we classified the following positions: 9 adjutants (adjutant), 3 bombardiers, 58 fusiliers (boschieters), 10 cadets, 14 captains, 7 captain lieutenants, 18 lieutenants, 2 lieutenant colonels, 1 colonel, 9 constables (of which 4 were constable mates), 173 corporals, 1 extraordinary lieutenant, 4 low ranking officers (landspassant), 3 garrison writers (guarnisoen schrijver), 1 garrison servant, 14 helpers (handlangers), 2 majors, 12 flutists (pijpers), 3 rapport writers (rapport schrijver), 117 sergeants, 1986 soldiers (soldaat), 4 swordsmen (swaartvegers), 37 drummers (tamboer), 4 trumpeters (trompetter), 42 ensigns (vaandrig), and 1 troop sergeant-major (wagmeester).

14 About a third of these merchants were originally from Amsterdam, the largest of the seven chambers that comprised the VOC.

Shipping

Shipping was important given the nature and purpose of the VOC's colony at the Cape of Good Hope (see, for example, Boshoff and Fourie, 2010), and unsurprisingly, the database has a large number (399) of entries for this sector, from 1699 to 1756. The associated occupations include 41 midshipmen (adelborst); 3 sailors (bootsgesel); 10 crew (bootzman), of which 3 were crew mates; 1 captain; 1 constable's mate (constapelsmaat); 1 equipment master (equipagie meester); 2 accountants; 292 sailors (mattroos); 4 assistant steersmen (onder stuurman), 2 head steersmen; 5 sailmakers (zijlmaker); 7 skippers (schipper); 27 quartermasters (quartiermeester), and 3 shipping corporals (scheepscorporal).

Trades

In addition to the occupations of skilled tradesman (such as butchers and bakers), this is also the residual category of the database where occupations are grouped that do not fit into the other categories. In total, the sector comprised 224 entries from 1699 to 1756. It included the following: 24 coopers, of which 5 were head coopers (kuijper); 1 head butcher (baas slaagter); 20 wagon makers, of which 3 were head wagon makers (wagenmakers); 9 bakers (bakkers); 3 bookbinders (boekbinder); 5 bottlers (bottelier); 2 truss makers (breukbandmaker); 1 gun smith (bussemaker); 2 map makers (caartemaaker); 3 painters (cladschilders); 7 cooks, of which 3 were cooks mates; 5 glaziers (glasemaker); 1 sail maker (haxelsnijder); 3 timber entry clerks (houtschrijvers); 4 gunstockmakers (lademakers); 1 surveyor (landmeter); 3 tanners (leerberijder); 6 millers, of which 1 was a servant (moolenaar); 1 watch maker (orlogiemaker); 2 pump makers (pompemakers); 1 pottery maker (pottebacker); 8 thatchers (rietdekkers); 9 gunmakers (roermakers); 1 seamstress (sakkenaijer); 2 butchers (seegelslagers); 2 locksmiths (slotemaker); 2 brick bakers (steen pannebacker); 4 tanner (velleberijder); 17 wagon makers (wagenmakers); 7 saddlers (zadelmaker); 4 head smiths; 4 tinsmiths (blikslager); 10 coppersmiths (kooperslager); 1 locksmith (slootemaker); and 1 smith (groffsmid).

Transport and storage

Finally, the extensive geographical spread of the colony ensured an important role for those who worked in the transport sector. We classified 220 entries in our database as relating to transport, including 138 wagon riders (wagenrijder, karrerijders), of which 2 were head wagon riders; 3 heads at the barn (baas aan de schuur); 4 coachmen (coetsier); and 65 stable hands (stalknegts), of which 8 were stable masters (baas aan de schuur, stalmeester).

APPENDIX 2

Meat

We have two price series for meat, one from Van Duin and Ross and one from the Scheepsoldijboeken. Van Duin and Ross (1987): Appendix 10, Table 2) provide the prices paid by the VOC for meat supplied to the Company for a number of years, starting in 1701 and ending in 1793. We created an annual time series from 1699 to 1793 by the linear interpolation of the data series and by assuming that the values for 1699 and 1700 were the same as for 1701. While these assumptions doubtless lead to inaccuracy at any particular point in time, they are unlikely to distort the long-run trend of the price series, which is the concern of this study. The series from the Scheepsoldijboeken is very similar to that of Van Duin and Ross, both in terms of levels and their variation over the century, with a correlation coefficient of 0.83 from 1701 to 1794. Both series show a striking decline in the price of meat over the course of the 18th century, by as much as 75% in the Van Duin and Ross

Sources and conversion formulae for price data

Item	Data Source	Unit in source data	Unit in consumption baskets	Conversion formulae
Grain	Scheepsoldijboeken	Mudde	Kg	111.5 litres per mud (De Zwart, 2011) and 0.76 litres per kilogram (Allen, 2001)
Bread	Scheepsoldijboeken	Pound	Kg	0.49409 kg per pound (De Zwart, 2011)
Beans	Scheepsoldijboeken	Mudde	Kg	111.5 litres per mud and 0.76 litres per kilogram (Allen, 2001)
Meat	Scheepsoldijboeken	Pound	Kg	0.49409 kg per pound
Butter	Scheepsoldijboeken	Pound	Kg	0.49409 kg per pound
Soap	Scheepsoldijboeken	Pound	Kg	0.49409 kg per pound
Candles	Scheepsoldijboeken	Pound	Kg	0.49409 kg per pound
Lamp oil	Scheepsoldijboeken	Container (Kan)	Litre	1 Kan equals 1.51 litres (Kooijmans and Schooneveld-Oosterling, 2000)
Cotton	Scheepsoldijboeken	Stuks Bafta	m ²	Price per m ² : $\frac{\text{BAFTA price}}{12.78 \times 0.93}$ (De Zwart, 2011)
Fuel	Scheepsoldijboeken	Vracht	MBTU	Price per MBTU is: $\frac{\text{Vracht price}}{1.58} \times 0.14$ (De Zwart, 2011)
Wine	Van Duin and Ross (1987)	Aam	Litres	1 Aam is 155.2 litres (Kooijmans and Schooneveld-Oosterling, 2000)

series and 80% in the *Scheepsoldijboeken*. At the start of the century, the price of meat was 0.35 guilders per kilogram in Amsterdam¹⁵ and 0.3 guilders in Cape Town, but by 1794 the price had fallen to 0.06 guilders in Cape Town and had risen to 0.42 guilders in Amsterdam. We used the price series from the *Scheepsoldijboeken* in the index calculations.

Grain and bread

Van Duin and Ross (1987), the auction lists from estates at the Cape Colony (now available in the TEPC database), as well the *Scheepsoldijboeken* provide price data for grain (tarwe). Both the Van Duin and Ross (1987) data and the auction lists are necessary to create a price series from 1700 until 1793, with the auction lists providing the entries prior to 1749, the starting period of the Van Duin and Ross (1987) series. We recorded all prices for grain over the period 1685 to 1775 from the auctions, taking the median value for years with more than one entry. Where the data overlaps with the VOC data (during the 1760s and early 1770s), the auction prices are lower than the VOC prices, as expected¹⁶. To merge these data into a single price series for the period 1699 to 1793, we used the auction prices as the base until 1749, and Van Duin and Ross's data from 1750 onwards. The auction prices for 1749 were used as the base for a new price series, together with the period-by-period changes in the Van Duin and Ross data, starting with the auction prices of 1749. The adjusted Van Duin and Ross data were then added to the auction data to form a single price series running to 1793¹⁷.

A further complication with this combined series is the inclusion of a small quantity of pulses in the export volumes, from which we derived the grain prices for the latter half of the 18th century. While the inaccuracy is not likely to be large, since pulses rarely accounted for more than 5% of exports, this figure was typically closer to 1–2% of the total export volume recorded in this series. Hence, this inaccuracy combined with the need to splice the data mid-century accounts for the decision to use the grain data from the *Scheepsoldijboeken*.

During this period, grain was typically consumed as a staple in the form of bread. However, De Zwart calculated his consumption basket with grain, and in order to compare his work, on the one hand, with that of Allen and others (2011), on the other, who used bread, we calculated consumption baskets alternatively

15 Van Zanden: "The prices of the most important consumer goods, and indices of wages and the cost of living in the western part of the Netherlands, 1450-180": see <http://www.iisg.nl/hpw/brenv.php>

16 The lower prices fetched for grain at auctions reflects (1) the premium paid by the VOC for grain delivered to Cape Town and (2) the transport cost from the countryside to Cape Town.

17 The unusually high auction value for 1708 (nearly twice as high as neighbouring values) created additional difficulties. This value will greatly affect the base value of the series in 1700, and if incorrect will exaggerate the decline in the grain price index during the course of the 18th century.

with bread and grain. The bread price series in the *Scheepsoldijboeken* used here is for hard bread (*hardebood*, made from flour, salt and water) a major component of soldier rations in this period. The price series for bread and grain are highly correlated over the course of the century (with a correlation coefficient of 0.87), though the cumulative decline in the price of bread (around 38%) exceeded that of grain (around 31%).

Wine

Van Duin and Ross (1987: Appendix 5, Tables 1 and 4) provide export data (values and volumes at an annual frequency) for the export of wine by the VOC from the Cape for the period 1749 to 1793. This data could be used to calculate an annual price series over the relevant sample. Meanwhile, auction data is available for the earlier half of the 18th century and was used to trace the price of wine from the late 17th century onwards and we used the median value for years with more than one entry.

To create a single series, we spliced the Van Duin and Ross data with the data for the auction in 1747, using the auction data's level in that year as the base level for the Van Duin and Ross data from that point onwards. Wine auctions recorded volatile wine prices during the first half of the century, against an upward trend, rising by about 60% over the first half of the 18th century. Prices stabilised from that point onwards, and the cumulative rise in wine prices between 1700 and 1794 was 57%.

Beans

The *Scheepsoldijboeken* provides a complete list of the prices of beans sold to ships passing the Cape of Good Hope for the years between 1706 and 1794. Price data is only available from 1706 onwards, and assuming the same price for 1706, values were filled in for 1700 to 1705. The price of beans shows a steady decline throughout the period, leaving beans 58% cheaper in 1794 than in 1706.

Butter

The price of butter was also recorded in the *Scheepsoldijboeken*, though the series was patchy, with entries for the following years only: 1708, 1716, 1717, 1720, 1741, 1748, 1760-1764, 1769, 1783, 1784, 1786, 1789-1791 and 1793. The series was completed through a process of linear interpolation, and the price for 1700 was assumed to be the same as for 1708. The price for butter rose sharply at the start of the century, nearly tripling by 1714, but then declined again to end 1794 at a level not very different from the price in 1708.

Soap

The price series for soap had to be constructed from three different data sources. The auction data has entries for 1708, 1714, 1751, 1752, 1755, 1756, and 1772. Two more data points (1716 and 1718) were collected from the *Scheepssoldijboeken*, while data for 1747-1750 and 1775 were collected from 'Prices Rendementen verkoop' (Nationaal Archief, (ref. no.) 1.04.02).¹⁸ It is assumed that the price of soap for 1700 was the same as for 1708. The price of soap varied considerably over the course of the 18th century, and by 1794 was 190% more expensive than in 1708.

Cotton

Since textiles are less homogenous than many of the other commodities discussed here, the construction of this price series presented specific challenges. While the auction lists record pieces of linen (*stukke linne*) of different kinds, they rarely show enough information to control for quality, size and age. As a result, the price series for cotton recorded in the *Scheepssoldijboeken* was the only practical source. This series showed at least three large cycles of price rises and declines during the course of the 18th century and items were 40% cheaper by 1794 than they were in 1708 (the first entry in the *Scheepssoldijboeken*).

Fuel

Two fuels for lighting (candles and lamp oil) and one for heating (firewood) are typically included in consumption baskets for this period (Allen et al. 2011). The *Scheepssoldijboeken* recorded data for all three. Starting with the price series for lamp oil, in addition to two pronounced price spikes early in the 18th century, the price of lamp oil rose steadily from the middle of the century onwards and the price of lamp oil was twice as expensive from the 1770s onwards than it had been in 1700. The price of candles (wax kaarsen) rose modestly (by about 25% cumulatively) in the second and third quarters of the 18th century, only to decline somewhat in the fourth quarter, prior a sharper price increase during the 1790s, which left candles 37% more expensive in 1794 than in 1700.

18 We are grateful to Pim de Zwart for kindly sharing this data with us.